Database Design Specification

for

VisionEdge: AI-Powered Optical Store Management & Virtual Try-On System

Version 1.0

Database Engine: MySQL 8.0 / MariaDB (Compatible with SQLite 3)

ORM Framework: SQLAlchemy 2.0 (Flask-SQLAlchemy)

Date: August 2026

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1. Executive Summary & Architectural Overview

The VisionEdge database architecture provides a robust relational data store supporting clinical optometry workflows, artificial intelligence vision scanning, interactive Virtual Try-On (VTO), inventory management, order processing, and GST-compliant invoicing.

The schema is built on MySQL 8.0 / MariaDB using the InnoDB storage engine for ACID transactional compliance and foreign key enforcement, while maintaining multi-database compatibility with SQLite 3 via Python's Flask-SQLAlchemy ORM.

Key Architectural Characteristics:

- Storage Engine: InnoDB with utf8mb4 character encoding (full Unicode support).

- Primary Key Strategy: Auto-incrementing 32-bit Integers for high-performance join operations, supplemented by unique alphanumeric business keys (customer_code, prescription_number, order_id, invoice_number).
- Transactional Guarantee: ACID compliance with automated rollback on failed order/billing transactions.
- Integrity Enforcement: Explicit Foreign Key constraints utilizing ON DELETE CASCADE for parent-child dependencies (e.g., customer -> customer_profile) and ON DELETE SET NULL for historical audit references.

2. Entity-Relationship (ER) Visual Model

The diagram below illustrates the core entities and their relationships within the VisionEdge ecosystem:

```
erDiagram
USER ||--o{ CUSTOMER : "creates"
USER ||--o{ AUDIT_LOG : "generates"
USER ||--o{ CHAT_HISTORY : "sends"

CUSTOMER ||--|| CUSTOMER_PROFILE : "has"
CUSTOMER ||--o{ APPOINTMENT : "schedules"
CUSTOMER ||--o{ EYE_TEST : "undergoes"
CUSTOMER ||--o{ PRESCRIPTION : "receives"
CUSTOMER ||--o{ ORDER : "places"
CUSTOMER ||--o{ FRAME_RECOMMENDATION : "receives"
CUSTOMER ||--o{ VIRTUAL_TRY_ON_HISTORY : "saves"

APPOINTMENT ||--o| EYE_TEST : "leads to"
EYE_TEST ||--o| PRESCRIPTION : "generates"

PRESCRIPTION ||--o{ ORDER : "applied to"
PRESCRIPTION ||--o{ LENS_RECOMMENDATION : "drives"

FRAME ||--o{ ORDER_ITEM : "included in"
FRAME ||--o{ VIRTUAL_TRY_ON_HISTORY : "previewed in"
LENS ||--o{ ORDER_ITEM : "included in"

ORDER ||--|{ ORDER_ITEM : "contains"
ORDER ||--|| BILLING : "billed via"
ORDER ||--o{ ORDER_STATUS_HISTORY : "tracks"

BILLING ||--o{ PAYMENT : "settled by"
```

3. Database Conceptual & Logical Modules

The schema is divided into 5 logical domain modules:

3.1 Module 1: User Authentication & Administration

Handles system login accounts, role-based security assignments (admin, staff, customer), system notifications, AI assistant logs, and security audit trails.

- Tables: user, audit_log, notification, chat_history.

3.2 Module 2: Customer CRM & Clinical Eye Examinations

Manages customer demographics, clinical medical notes (eye conditions, blood group, allergies), appointment schedules, OD/OS clinical eye tests, and AI Gemini OCR scanned prescriptions.

- Tables: customer, customer_profile, appointment, eye_test, prescription.

3.3 Module 3: Frames & Optical Lens Inventory

Controls spectacle frame attributes (shape, brand, material, size, price, stock, transparent VTO PNG URL), lens specifications (design type, refractive coating, price, stock), supplier contacts, and inventory stock adjustment ledgers.

- Tables: frame, lens, supplier, brand, category, inventory_transaction.

3.4 Module 4: Sales Orders, Invoicing & Payment Processing

Manages retail sales orders, itemized order lines (frames + lenses), GST tax calculations, payment modes (Cash, Card, UPI), billing receipts, and order fulfillment states.

- Tables: order, order_item, billing, payment, order_status_history.

3.5 Module 5: AI Vision, Recommendations & Virtual Try-On

Stores algorithmic recommendations matching frame geometry to face shapes, clinical prescription lens recommendations, and saved customer Virtual Try-On snapshot images.

- Tables: frame_recommendation, lens_recommendation, virtual_try_on_history.

4. Complete Data Dictionary (23 Tables)

4.1. Table: `user`

Manages system authentication credentials, user profiles, and authorization roles.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Unique user ID
username	VARCHAR(80)	NOT NULL, UNIQUE	*None*	Account login handle
first_name	VARCHAR(50)	NULLABLE	NULL	Given first name
last_name	VARCHAR(50)	NULLABLE	NULL	User surname
password_hash	VARCHAR(255)	NOT NULL	*None*	Bcrypt/PBKDF2 secure hashed password
email	VARCHAR(120)	UNIQUE, NULLABLE	NULL	Contact email address
phone	VARCHAR(20)	NULLABLE	NULL	Phone number
gender	VARCHAR(20)	NULLABLE	NULL	Gender identification
date_of_birth	DATE	NULLABLE	NULL	Date of birth
address	TEXT	NULLABLE	NULL	Street address
role	VARCHAR(50)	NOT NULL	'customer'	Security role (admin, staff, customer)
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Account creation timestamp

4.2. Table: `customer`

Stores demographic data and contact information for store clients.

Column Name	Data Type	Constraints	Default Value	Description

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Unique customer ID
customer_code	VARCHAR(20)	UNIQUE	NULL	Alphanumeric business code (e.g. CUST-101)
first_name	VARCHAR(50)	NOT NULL	*None*	Customer first name
last_name	VARCHAR(50)	NOT NULL	*None*	Customer last name
gender	VARCHAR(20)	NULLABLE	NULL	Customer gender
date_of_birth	DATE	NULLABLE	NULL	Birthdate
age	INT	NULLABLE	NULL	Calculated age in years
phone	VARCHAR(20)	NOT NULL, INDEXED	*None*	Primary telephone number
alternate_phone	VARCHAR(20)	NULLABLE	NULL	Secondary phone number
email	VARCHAR(120)	INDEXED, NULLABLE	NULL	Email address
address	TEXT	NULLABLE	NULL	Residential address
city	VARCHAR(100)	NULLABLE	NULL	Resident city
state	VARCHAR(100)	NULLABLE	NULL	Resident state
pincode	VARCHAR(20)	NULLABLE	NULL	Postal pincode
occupation	VARCHAR(100)	NULLABLE	NULL	Profession/Occupation
photo	VARCHAR(255)	NULLABLE	NULL	Path to customer profile image
status	VARCHAR(20)	NOT NULL	'Active'	Customer status (Active, Inactive)
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Profile creation timestamp
updated_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Last update timestamp
created_by	INT	FOREIGN KEY (user.id)	NULL	Staff user who registered customer

4.3. Table: `customer_profile`

Stores extended medical notes and clinical parameters (1-to-1 extension of customer).

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Profile record ID
customer_id	INT	UNIQUE, FK (customer.id)	*None*	Linked customer ID (ON DELETE CASCADE)
blood_group	VARCHAR(10)	NULLABLE	NULL	Blood group (e.g. O+, A-)
eye_conditions	TEXT	NULLABLE	NULL	Known eye conditions (e.g., Myopia, Cataract)
allergies	TEXT	NULLABLE	NULL	Known allergies
notes	TEXT	NULLABLE	NULL	Additional clinical observations

4.4. Table: `appointment`

Tracks customer visits and consultation schedules.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Appointment ID

Column Name	Data Type	Constraints	Default Value	Description
customer_id	INT	FK (customer.id)	*None*	Customer ID (ON DELETE CASCADE)
appointment_date	DATE	NOT NULL	*None*	Scheduled visit date
appointment_time	TIME	NOT NULL	*None*	Scheduled visit time
appointment_type	VARCHAR(50)	NOT NULL	'Eye Test'	Type (Eye Test, Frame Trial, Consultation)
doctor_name	VARCHAR(100)	NULLABLE	NULL	Assigned doctor/optometrist name
status	VARCHAR(50)	NOT NULL	'Pending'	Status (Pending, Confirmed, Completed, Cancelled)
notes	TEXT	NULLABLE	NULL	Appointment notes
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Entry creation timestamp

4.5. Table: `eye_test`

Stores raw numerical clinical eye measurements recorded during an eye exam.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Eye exam ID
customer_id	INT	FK (customer.id)	*None*	Linked customer (ON DELETE CASCADE)
appointment_id	INT	FK (appointment.id)	NULL	Linked appointment (ON DELETE SET NULL)
right_sphere	VARCHAR(20)	NULLABLE	NULL	OD Sphere (SPH) power
right_cylinder	VARCHAR(20)	NULLABLE	NULL	OD Cylinder (CYL) power
right_axis	VARCHAR(20)	NULLABLE	NULL	OD Axis orientation angle
left_sphere	VARCHAR(20)	NULLABLE	NULL	OS Sphere (SPH) power
left_cylinder	VARCHAR(20)	NULLABLE	NULL	OS Cylinder (CYL) power
left_axis	VARCHAR(20)	NULLABLE	NULL	OS Axis orientation angle
pd_value	VARCHAR(20)	NULLABLE	NULL	Pupillary Distance (PD in mm)
remarks	TEXT	NULLABLE	NULL	Optometrist remarks
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Examination timestamp

4.6. Table: `prescription`

Stores digital prescription details and AI Gemini OCR parsing results.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Prescription ID
prescription_number	VARCHAR(50)	UNIQUE, NULLABLE	NULL	Unique tracking code (e.g. RX-2026-04)
customer_id	INT	FK (customer.id)	*None*	Customer ID (ON DELETE CASCADE)
eye_test_id	INT	FK (eye_test.id)	NULL	Clinical exam ID (ON DELETE SET NULL)
add_power	VARCHAR(20)	NULLABLE	NULL	Near vision Add power
prescription_image	VARCHAR(255)	NULLABLE	NULL	File path of uploaded prescription scan
ai_extracted_text	TEXT	NULLABLE	NULL	Raw JSON text extracted by Gemini AI
confidence_score	FLOAT	NULLABLE	NULL	Gemini AI extraction confidence (0.0 to 1.0)

Column Name	Data Type	Constraints	Default Value	Description
doctor_name	VARCHAR(100)	NULLABLE	NULL	Prescribing optometrist name
date	DATE	NULLABLE	NULL	Issue date
expiry_date	DATE	NULLABLE	NULL	Expiry date
status	VARCHAR(20)	NOT NULL	'Completed'	Prescription status
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Record timestamp

4.7. Table: `frame`

Stores spectacle frame inventory attributes, stock levels, and Virtual Try-On PNG image URLs.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Frame ID
frame_code	VARCHAR(50)	UNIQUE, NULLABLE	NULL	Frame SKU code (e.g., FRM-001)
brand	VARCHAR(100)	NULLABLE	NULL	Brand manufacturer (e.g. Ray-Ban)
model	VARCHAR(100)	NULLABLE	NULL	Frame model designation
name	VARCHAR(150)	NULLABLE	NULL	Display title
shape	VARCHAR(50)	NULLABLE	NULL	Geometry (Square, Oval, Round, Cat-Eye)
color	VARCHAR(50)	NULLABLE	NULL	Frame color
size	VARCHAR(50)	NULLABLE	NULL	Frame size (Small, Medium, Large)
material	VARCHAR(50)	NULLABLE	NULL	Material (Acetate, Titanium, Metal)
gender	VARCHAR(20)	NULLABLE	NULL	Gender target (Men, Women, Unisex)
price	FLOAT	NOT NULL	0.0	Retail price in INR
selling_price	FLOAT	NULLABLE	NULL	Discounted selling price
stock	INT	NOT NULL	0	Available stock count
min_stock	INT	NOT NULL	5	Minimum stock threshold alert level
image	VARCHAR(255)	NULLABLE	NULL	Primary product image URL
vto_image	VARCHAR(255)	NULLABLE	NULL	Transparent PNG URL for Virtual Try-On overlay
status	VARCHAR(50)	NOT NULL	'Available'	Status (Available, Out of Stock)
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Insertion timestamp
updated_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Last update timestamp

4.8. Table: `lens`

Stores optical lens inventory attributes, design types, and specialty coatings.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Lens ID
lens_code	VARCHAR(50)	UNIQUE, NULLABLE	NULL	Lens SKU code (e.g. LNS-101)
name	VARCHAR(150)	NULLABLE	NULL	Lens product title
brand	VARCHAR(100)	NULLABLE	NULL	Lens manufacturer (e.g. Essilor, Zeiss)

Column Name	Data Type	Constraints	Default Value	Description
type	VARCHAR(50)	NULLABLE	NULL	Design (Single Vision, Progressive, Bifocal)
coating	VARCHAR(100)	NULLABLE	NULL	Coating (Anti-Glare, Blue Cut, Photochromic)
price	FLOAT	NOT NULL	0.0	Retail price in INR
stock	INT	NOT NULL	0	Available lens stock count
min_stock	INT	NOT NULL	10	Low stock alert threshold
image	VARCHAR(255)	NULLABLE	NULL	Lens product image URL
status	VARCHAR(50)	NOT NULL	'Available'	Status (Available, Out of Stock)
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Record creation timestamp

4.9. Table: `supplier`

Stores vendor contact details for inventory restocking.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Supplier ID
name	VARCHAR(150)	NOT NULL	*None*	Vendor / Business name
phone	VARCHAR(20)	NULLABLE	NULL	Contact phone number
email	VARCHAR(120)	NULLABLE	NULL	Contact email
address	TEXT	NULLABLE	NULL	Supplier physical address
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Entry creation timestamp

4.10. Table: `order`

Stores sales order header information.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Order ID
order_id	VARCHAR(50)	UNIQUE, NULLABLE	NULL	Customer-facing order code (e.g., ORD-1001)
customer_id	INT	FK (customer.id)	*None*	Purchasing customer (ON DELETE CASCADE)
prescription_id	INT	FK (prescription.id)	NULL	Linked prescription (ON DELETE SET NULL)
status	VARCHAR(50)	NOT NULL	'Pending'	Order stage (Pending, Processing, Ready, Delivered)
total_amount	FLOAT	NOT NULL	0.0	Gross order total amount
order_date	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Date of placement
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Record creation timestamp

4.11. Table: `order_item`

Itemized line entries (frames and lenses) associated with a sales order.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Line item ID
order_id	INT	FK (order.id)	*None*	Parent order reference (ON DELETE CASCADE)
product_type	VARCHAR(50)	NULLABLE	NULL	Product classification (Frame, Lens)
product_id	INT	NULLABLE	NULL	Foreign key referencing frame.id or lens.id
quantity	INT	NOT NULL	1	Purchased quantity
price	FLOAT	NOT NULL	0.0	Unit price at time of purchase
total_price	FLOAT	NOT NULL	0.0	Total line item price (quantity * price)

4.12. Table: `billing`

Stores tax breakdown, payment method, and PDF invoice receipt references.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Billing record ID
invoice_number	VARCHAR(50)	NOT NULL, UNIQUE	*None*	Legal invoice number (e.g. INV-2026-001)
order_id	INT	FK (order.id)	*None*	Associated sales order (ON DELETE CASCADE)
subtotal	FLOAT	NOT NULL	0.0	Amount before tax
discount	FLOAT	NOT NULL	0.0	Total promotional discount applied
gst	FLOAT	NOT NULL	0.0	Goods and Services Tax amount (18%)
total_amount	FLOAT	NOT NULL	0.0	Net payable invoice total
payment_status	VARCHAR(20)	NOT NULL	'Pending'	Settlement state (Paid, Pending)
payment_method	VARCHAR(50)	NULLABLE	'Cash'	Payment mode (Cash, Card, UPI)
invoice_pdf	VARCHAR(255)	NULLABLE	NULL	File path to generated PDF receipt
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Invoice creation timestamp

4.13. Table: `payment`

Detailed ledger tracking payment transactions per invoice.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Transaction ID
billing_id	INT	FK (billing.id)	*None*	Parent billing record (ON DELETE CASCADE)
amount	FLOAT	NOT NULL	0.0	Transaction monetary amount
transaction_id	VARCHAR(100)	NULLABLE	NULL	POS transaction/reference ID
payment_method	VARCHAR(50)	NULLABLE	NULL	Method used
payment_date	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Transaction timestamp

4.14. Table: `frame_recommendation`

Stores face shape detection results and AI frame geometry recommendations.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Recommendation ID
customer_id	INT	FK (customer.id)	*None*	Linked customer (ON DELETE CASCADE)
face_shape	VARCHAR(50)	NULLABLE	NULL	Detected shape (Oval, Round, Square, Heart)
recommended_frames	TEXT	NULLABLE	NULL	JSON array of recommended frame IDs
selected_frame	VARCHAR(100)	NULLABLE	NULL	Customer selected frame
ai_reason	TEXT	NULLABLE	NULL	Algorithmic matching explanation
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Recommendation timestamp

4.15. Table: `lens_recommendation`

Stores AI prescription lens recommendations based on diopter severity and usage profile.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Recommendation ID
customer_id	INT	FK (customer.id)	NULL	Customer ID (ON DELETE SET NULL)
prescription_id	INT	FK (prescription.id)	NULL	Prescription ID (ON DELETE SET NULL)
recommended_lens	TEXT	NULLABLE	NULL	Recommended lens type & index
reason	TEXT	NULLABLE	NULL	Clinical logic explanation
ai_response	TEXT	NULLABLE	NULL	Raw AI model response
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Timestamp

4.16. Table: `virtual_try_on_history`

Stores customer Virtual Try-On snapshots and frame overlays.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	History record ID
customer_id	INT	FK (customer.id)	*None*	Customer ID (ON DELETE CASCADE)
frame_id	INT	FK (frame.id)	*None*	Tried-on frame ID (ON DELETE CASCADE)
tryon_image	VARCHAR(255)	NULLABLE	NULL	Snapshot image URL
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Snapshot capture timestamp

4.17. Table: `chat_history`

Logs customer queries and AI Optical Assistant chat responses.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Chat log ID

Column Name	Data Type	Constraints	Default Value	Description
user_id	INT	FK (user.id)	NULL	User ID (ON DELETE SET NULL)
customer_id	INT	FK (customer.id)	NULL	Customer ID (ON DELETE SET NULL)
question	TEXT	NOT NULL	*None*	User question string
answer	TEXT	NOT NULL	*None*	AI chatbot answer string
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Log timestamp

4.18. Table: `notification`

System alerts for low inventory stock, appointment reminders, and order ready updates.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Notification ID
user_id	INT	FK (user.id)	NULL	Target user (ON DELETE SET NULL)
customer_id	INT	FK (customer.id)	NULL	Target customer (ON DELETE SET NULL)
notification_type	VARCHAR(50)	NOT NULL	*None*	Type (Low Stock, Order Ready)
title	VARCHAR(150)	NOT NULL	*None*	Alert header title
message	TEXT	NOT NULL	*None*	Full message text
status	VARCHAR(20)	NOT NULL	'Unread'	Read status (Unread, Read)
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Alert creation timestamp

4.19. Table: `inventory_transaction`

Audit log tracking manual stock adjustments, supplier restocking, and sale deductions.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Transaction ID
product_type	VARCHAR(50)	NULLABLE	NULL	Product type (Frame, Lens)
product_id	INT	NULLABLE	NULL	Product item ID
transaction_type	VARCHAR(50)	NULLABLE	NULL	Type (Stock In, Stock Out, Adjustment)
quantity	INT	NOT NULL	0	Quantity change amount
notes	TEXT	NULLABLE	NULL	Movement reason notes
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Ledger timestamp

4.20. Table: `brand`

Reference catalog for optical brand manufacturers.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Brand ID

Column Name	Data Type	Constraints	Default Value	Description
name	VARCHAR(100)	NOT NULL	*None*	Brand name

4.21. Table: `category`

Reference catalog for product category types.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Category ID
name	VARCHAR(100)	NOT NULL	*None*	Category name

4.22. Table: `order_status_history`

Tracks timestamped status changes throughout an order's lifecycle.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Audit ID
order_id	INT	FK (order.id)	*None*	Target order (ON DELETE CASCADE)
status	VARCHAR(50)	NULLABLE	NULL	New status state
notes	TEXT	NULLABLE	NULL	Status transition notes
created_at	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Change timestamp

4.23. Table: `audit_log`

Security audit trail recording administrative and user actions across modules.

Column Name	Data Type	Constraints	Default Value	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	*None*	Log ID
user_id	INT	FK (user.id)	NULL	Actor user ID (ON DELETE SET NULL)
action	VARCHAR(100)	NULLABLE	NULL	Action performed (e.g. UPDATE_PRICE)
module	VARCHAR(100)	NULLABLE	NULL	Affected module (Inventory, Billing)
timestamp	DATETIME	NOT NULL	CURRENT_TIMESTAMP	Operation timestamp

5. Normalization & Relational Integrity Analysis

The VisionEdge database design adheres strictly to Third Normal Form (3NF):

1. First Normal Form (1NF):

- All columns contain atomic, scalar values (e.g., separate columns for first_name, last_name, right_sphere, left_sphere).
- Multi-valued arrays (such as order items or recommended frames) are decoupled into distinct child tables (order_item) or structured JSON fields.

2. Second Normal Form (2NF):

- Every non-key column in every table depends fully on the table's whole primary key.
- Compound attributes are properly decomposed into child tables (e.g., customer medical notes are separated into customer_profile).

3. Third Normal Form (3NF):

- Transitive dependencies are eliminated. For example, customer address and phone attributes reside purely in customer, avoiding duplication inside order or billing records. billing references order_id directly, deriving customer references through relational joins.

6. Indexing Strategy & Query Performance Optimization

To maintain API response times under 300ms, high-frequency lookup columns are indexed explicitly:

- Primary Indexes: Auto-incrementing B-Tree indexes on id for all 23 tables.
- Unique Secondary Indexes:
- user.username, user.email
- customer.customer_code
- prescription.prescription_number
- frame.frame_code, lens.lens_code
- order.order_id, billing.invoice_number
- Non-Unique Performance Indexes:
- idx_customer_phone: B-Tree index on customer(phone) for sub-millisecond front-desk customer lookup.
- idx_customer_email: B-Tree index on customer(email).
- Foreign key fields (customer_id, order_id, prescription_id, frame_id) are auto-indexed by MySQL InnoDB to eliminate full table scans during JOIN queries.

7. Foreign Key Cascade & Data Security Policies

Cascade Rules Strategy:

- ON DELETE CASCADE: Applied to tight parent-child structural relationships:
- Deleting a customer automatically removes their customer_profile, appointment, eye_test, prescription, frame_recommendation, and virtual_try_on_history.
- Deleting an order automatically removes its order_item, billing, and order_status_history entries.
- ON DELETE SET NULL: Applied to operational or financial historical logs where data preservation is required for accounting/audit compliance:
- prescription.eye_test_id -> ON DELETE SET NULL
- order.prescription_id -> ON DELETE SET NULL
- audit_log.user_id, chat_history.user_id -> ON DELETE SET NULL

Security Controls:

- Password hashes use secure cryptographic algorithms (Bcrypt / scrypt).
- Database connections utilize parameterized SQLAlchemy statements, rendering SQL injection impossible.

- Access control is enforced at the API layer via JWT role verification (admin, staff, customer).

End of Database Design Specification Document for VisionEdge.